

training input for LNN. LNN is trained to minimize the loss of logical contradictions. It learns symbolic rules from logical facts that can be mapped to action strategies, which are logical rules that can correspond to the best action strategies. Text-to-logic conversion is accomplished through a clear and explicit process, with the semantic parser explicitly mapping text information to a clear set of logical expressions. In some cases, the transformation process may involve enriching and supplementing these logical expressions with external knowledge bases, which generate more complex logical expressions based on associations between information in the text and data in the external knowledge base or consider previous observations or interaction history are used to construct the current logical state more accurately, so this process is also explicit. Likewise, these logical facts are logically reasoned and learned through LNN, operating based on clear, logical rules, so the logical decision-making process is explicit and explainable.

3.5 Unified Representation and Explicit Decision Making(prediction)

There are 3 Neuro-Symbolic AI studies in this category, with three commonalities. First, although they use neural networks to obtain features, the neural network's output maintains the exact representation that can be processed by symbolic logic. Second, the overall decision logic is fully explicit and interpretable.

Riegel et al. [149] proposed a LNN(logical neural network). As we mentioned in Section 3.4 of Part I, LNN directly interprets and operates logical operations by mapping each of its neurons to elements in the logical formula, which means that when the model processes information or makes decisions, its calculation process can be equivalent to performing a series of logical judgments, in which the output of each neuron not only represents the truth value of a logical proposition, and can also reflect how this truth value is derived from the input through logical operations, which is also the source of its high interpretability. In addition, each logical proposition in LNN is assigned a truth value range, and the uncertainty of the truth value of the proposition is captured and expressed through upper and lower bounds. For example, when faced with logical contradictions and incomplete knowledge, LNN will increase the uncertainty of the authenticity of certain propositions by expanding the truth value range of the proposition and finding a truth range adjustment solution that minimizes the contradiction. Similarly, when there is not enough information to determine the truth value of a proposition, LNN can reflect the uncertainty of the proposition by giving it a broader truth value range and dynamically adjust the true range as information is obtained to adapt to complex and dynamic changing information environment. Finally, unlike traditional single-task neural networks, LNN can simultaneously perform various logical reasoning tasks, such as theorem proving and fact derivation, demonstrating its high flexibility in logical processing and future development potential.

Sen et al. [160] proposed an LNN-based inductive logic programming method that learns interpretive rules from noisy, real-world data, which can generate interpretable logic rules as output based on structured input data. This method first uses a knowledge base containing facts and relations, as well as rules describing the form or structure of the target as input. Then, an LNN network is built based on the template to simulate logical connectives, where each node or neuron represents an expression or a logical rule combined with logical connectives. Because LNN allows the behavior of logical operations to be adjusted through the learning process, the facts in the knowledge base can be used as training data, and the parameters in the LNN can be correctly reflected through optimization algorithms such as back propagation and gradient descent, so that it can correctly reflect the logic relation between facts. Finally, the trained LNN can be converted into a set of logical rules that directly reflect the logical relationships in the input data. It can be used to reason, predict, or explain patterns in the data and is interpretable. This method applies LNN in the ILP direction, allowing it to handle the complexity and uncertainty in real-world data while improving learning efficiency and the quality of rules.

Sen et al. [159] proposed using LNN to complete the knowledge base. This method first defines the goal of knowledge base completion based on known entities and relationships. In this process, two methods, CM(Chain of Mixtures) and MP(Mixture of Paths), are mainly considered. Then, Boolean logic is extended to the real-valued domain in a parameterized manner through LNN, $LNN-\wedge$ is used to simulate the logical AND operation, $NN-pred$ is used to represent the relationship mixture or path mixture in the rule body, and LNN is used to construct knowledge base completion rules based on the rule template, that is, for the CM method, each hop relationship is represented as a mixture of relationships. In contrast, a mixture of multi-hop relationship sequences is directly learned for the MP method. This method uses pre-trained knowledge graph embeddings to supplement rule learning for the shared path sparsity problem in knowledge-based completion tasks. In this way, the parameters in the LNN can be optimized by iterative training on a given knowledge base graph, minimizing the prediction error. Finally, human-interpretable rules are extracted from the LNN model and used to predict missing entities in the knowledge base.

In addition to the above studies, Arrotta et al. [16, 17], He et al. [89], Xu et al. [197] proposed loss functions suitable for Neuro-Symbolic Learning, and [4] proposed a regularization method suitable for neuro-symbolic learning. However, the above studies did not improve the models' interpretability.

4 TRENDS

This study used the keywords 'neuro-symbolic', 'neuro symbolic', and 'neuro symbolic learning' to survey relevant research from 2014 to the present on Google Scholar and Research Gate. Although the search was conducted to the greatest extent, it is possible that some literature could not be included due to limitations. Therefore, the trend comparison below mainly aims to illustrate the evolution of research trends rather than provide an accurate total literature volume. According to the scope of this study, several apparent trends can be seen from 2014 to Feb of 2024: First, the number of papers has increased significantly, especially between 2020 and 2023; the number of papers published is on an upward trend, and its publication in 2023 has the most significant number of papers, 55 in total based on presenting data. These numbers reflect the primary growth in research interest and activity in neural symbolic systems.

Secondly, among current neuro-symbolic methods, images and text are the most common input data types, reflecting their ubiquity and importance in studying neuro-symbolic systems.

There are relatively few explorations into numerical and mathematical expression processing, structured data processing, environment and state awareness, and multimodal data types. However, as Figure 1 shows, the number of neuro-symbolic studies on these four input types has continued growing since 2016.

In addition, the vast majority of research used unimodal and non-heterogeneous representation space, which shows that in neuro-symbolic system research, a single type of data, such as text, image, or structured data, is still more common in research objects. Only a few papers explore the representation space of unimodal heterogeneous, multimodal non-heterogeneous, and multimodal heterogeneous, showing that these directions are relatively new and may be future growth points for research.

Finally, the most significant papers had medium-low explainability and low explainability. This figure indicates that most research results in this area still need to be higher in explainability. Only a few research have achieved medium, medium-high, and high explainability, indicating that there are still significant challenges in improving the explainability of neural symbolic systems in the future.

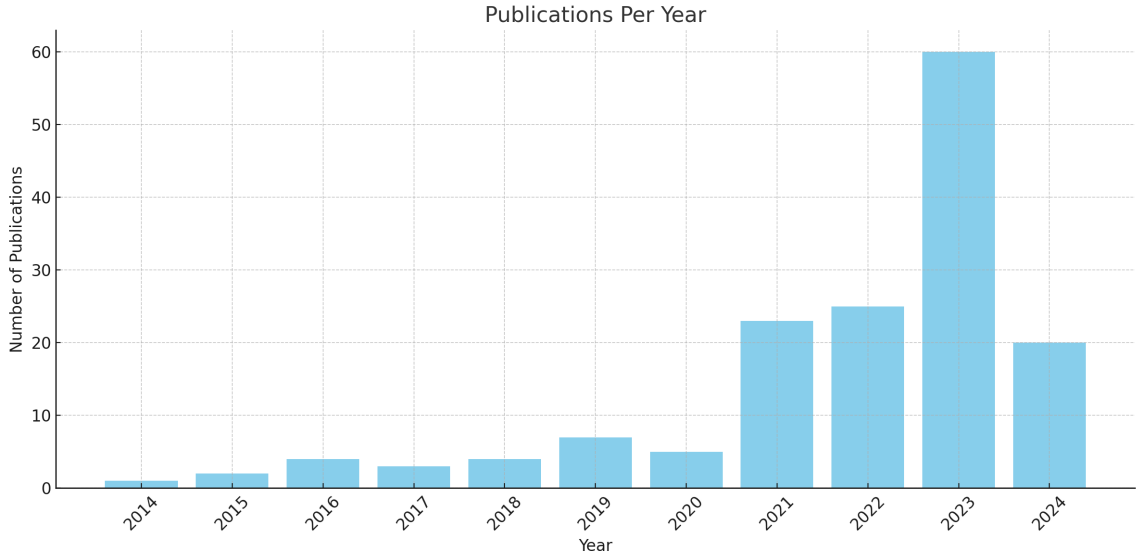


Fig. 1. Annual publication trends

5 MAJOR RESEARCH CHALLENGES

5.1 Unified Representations in Neural Networks and Symbolic Logic

The conversion of representation between neural networks and symbolic logic has constantly challenged neuro-symbolic learning. In the two most common situations of neuro-symbolic AI models, neural networks are utilized to augment the extraction of symbolic logical features by interpreting complex patterns or data structures, which enhances the effectiveness of traditional symbolic methods, or symbolic knowledge embedding is applied to provide rule constraints or reasoning logic for neural networks. However, these cooperations are usually optimized for specific tasks, which are difficult to adapt or transfer to new tasks or data sets. Therefore, retraining the neural network model or adjusting the symbolic logic rules are necessary when requirements change, which hints that specific combinations constrain the model's overall generalization ability. Alternatively, both diagrams can directly utilize the extracted features or learned knowledge when neural networks and symbolic logic modules use unified representation, bringing additional benefits, such as improving training and inference efficiency.

In addition, the two methods of cooperation between neural networks and symbolic logic mentioned above are synchronization processes of knowledge, except that they are inefficient and offline. However, applying the unified representation will make this synchronization more efficient. Specifically, this consistency avoids additional knowledge transformation steps and corresponding information loss, which can make the system more flexible and efficient, reducing reliance on large amounts of offline training data while decreasing the complexity of model update and maintenance in handling complex practical application problems such as multi-step reasoning tasks in natural language or image recognition.

Despite the benefits, designing an ideal unified representation still faces many obstacles. An ideal representation could capture the structural properties of symbolic logic while maintaining the essential patterns of the data. Such designs require a solid understanding of the data distribution and its latent relationship with the logical entities. For